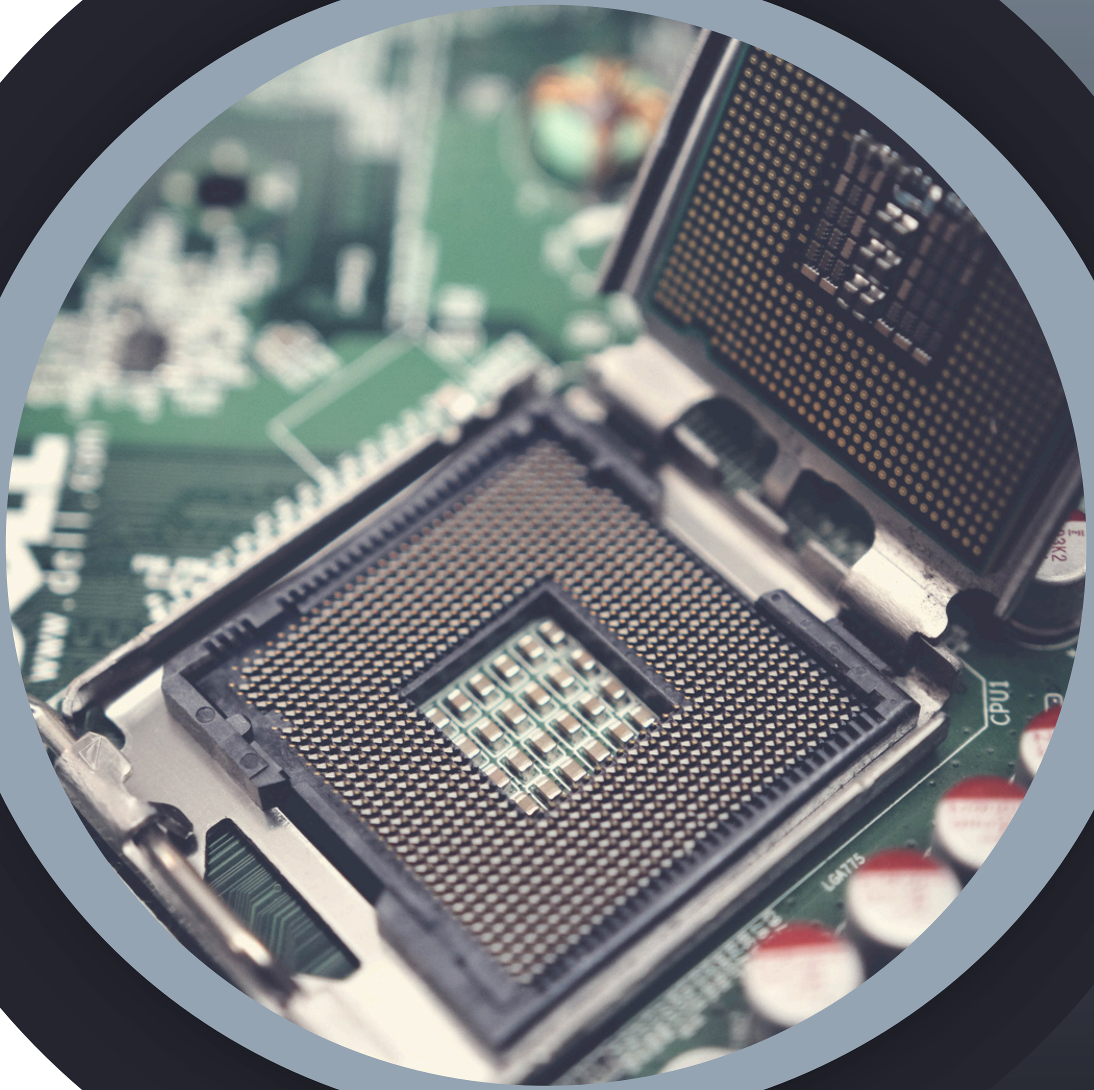


Hand Gesture-Based Access Control System with Single Board Computers

Importance of Gesture-based Systems

It is a hands-free way to interact with devices. It’s easier, and faster, also helps prevent germs from spreading, making it great for hospitals, factories, and homes.

It provides a user-friendly way for people with mobility impairments to interact with devices.

Applications

Healthcare Settings

• allows healthcare professionals to use equipment without touching surfaces, reducing germ spread and improving hygiene.

Smart Homes

•Control lights, fans, and other appliances with simple hand gestures, creating a seamless and modern home automation experience.

Secure Access Control

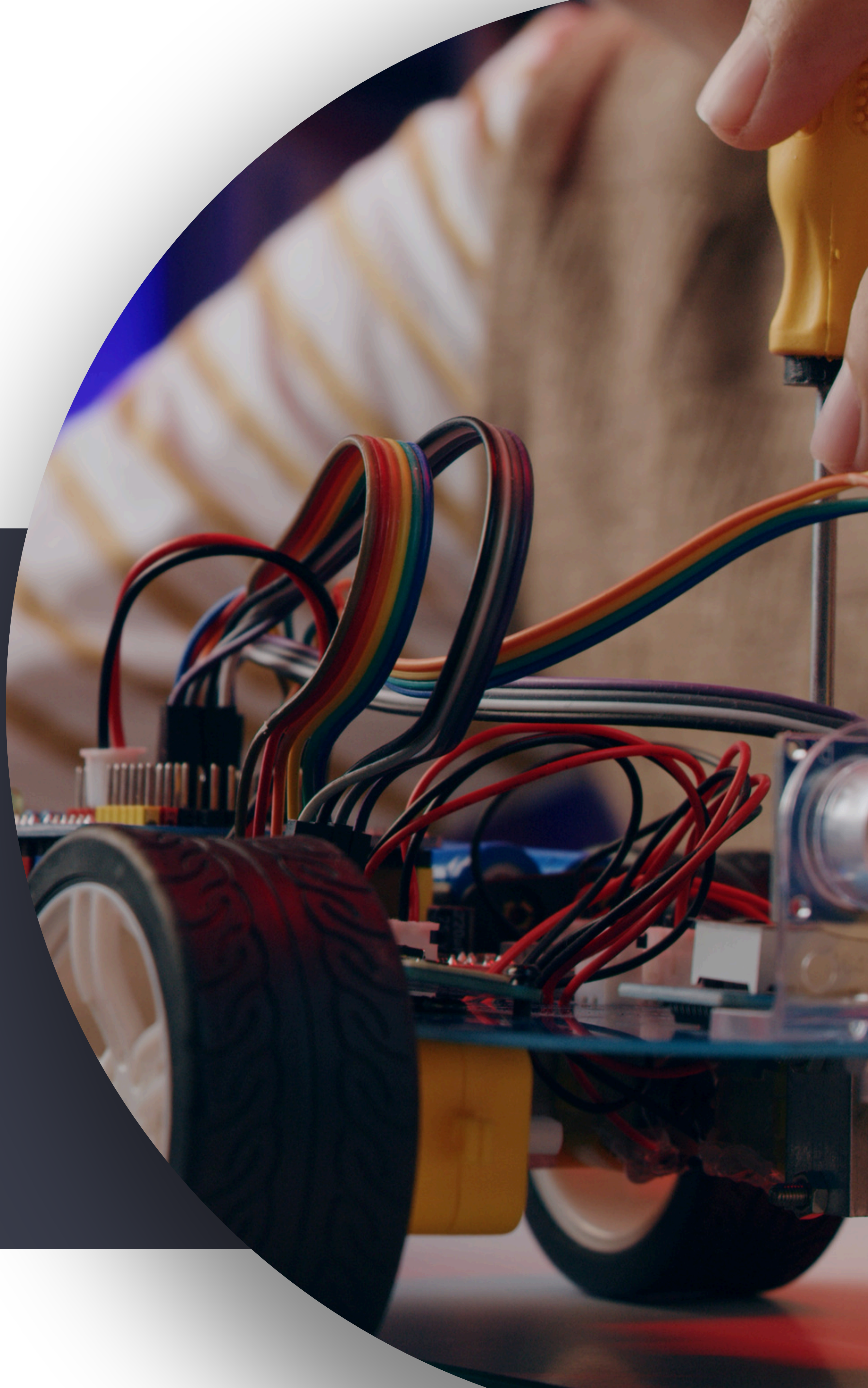
•Use hand gestures to unlock doors or access restricted areas in secure facilities, eliminating the need for keys or cards.

Industrial Automation

•Workers can control machinery without physical contact, enhancing safety and efficiency in environments where hands-free operation is crucial.

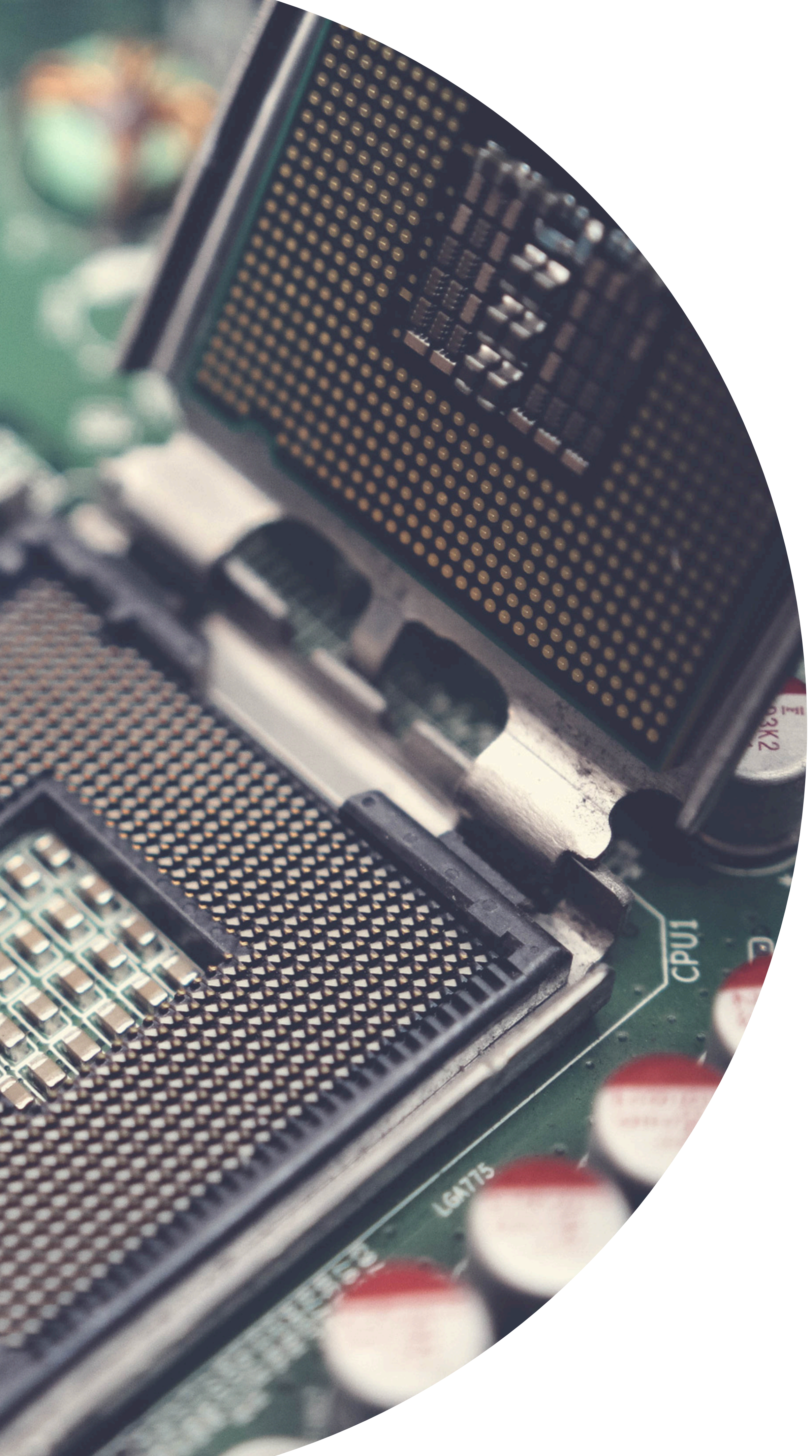
Overview

The hand gesture-based home automation system offers a touch-free, intuitive way to control appliances like lights and fans and monitor room temperature. Using computer vision and machine learning, enhances convenience, accessibility, and energy efficiency in smart homes by interpreting hand gestures to execute commands.



OBJECTIVES

1. **Intuitive Control**
   Users can control home appliances like lights and fans with simple hand gestures, making it easy for people of all ages to use.
2. **Energy Efficiency**
   The system smartly automates to reduce energy consumption by ensuring appliances are only on when needed.
3. **Touchless Interaction**
   Minimize physical contact with switches and remotes, reducing germ spread and enhancing hygiene
4. **Real-Time Monitoring**
   A specific hand gesture measures and displays room temperature, adding environmental awareness.
5. **Scalability**
   The system is designed to be scalable, allowing for future upgrades and integration with additional smart home devices



Hardware Components

Single Board Computer (SBC): Raspberry Pi or Jetson Nano for real-time video processing, gesture recognition, and device control.

Camera Module: High-definition camera (e.g., Raspberry Pi Camera Module or USB Webcam) to capture hand gestures.

Relay Module: Controls the on/off states of lights, fans, and other appliances.

Temperature Sensor: DHT11 or DHT22 to measure room temperature, triggered by specific hand gestures.

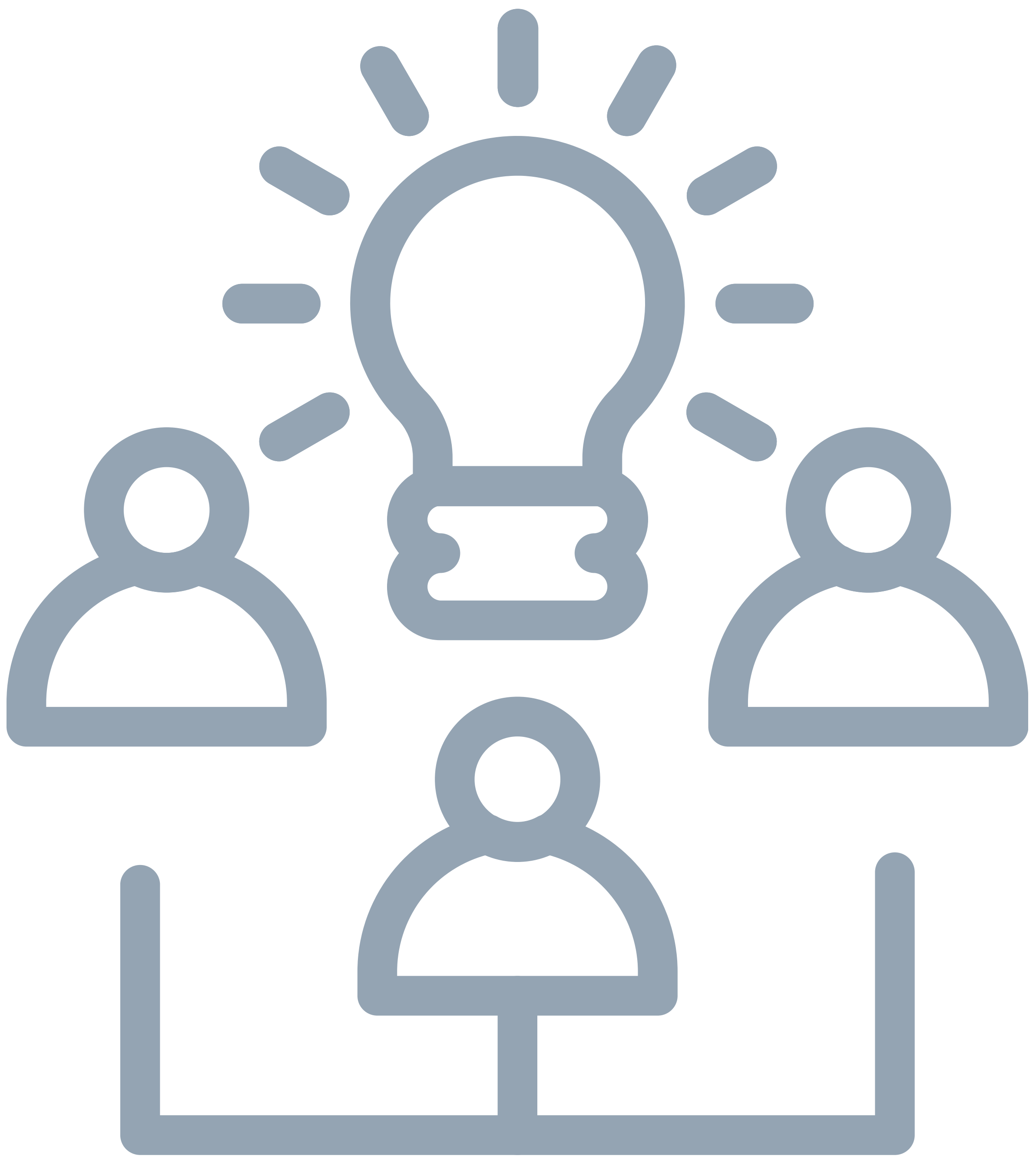
Power Supply: Provides stable power to the SBC and connected components.

Display Module: LCD or OLED screen to show room temperature or system statuses.

Actuators: For controlling additional devices or interfacing with the relay module, if needed.

Enclosure: Protective casing for the SBC, camera, and other electronics to ensure durability and safety.

Functional Requirement



- ✓ Gesture Recognition for Appliance Control
- ✓ Real-Time Room Temperature Measurement
- ✓ Appliance State Management
- ✓ System Configuration and Error Handling
- ✓ User Feedback

Non - Functional Requirement

- Accuracy
- Response Time
- Scalability
- Reliability
- Usability
- Security
- Maintainability
- Power Efficiency
- Compatibility

Symbols which we used

Device “ON”

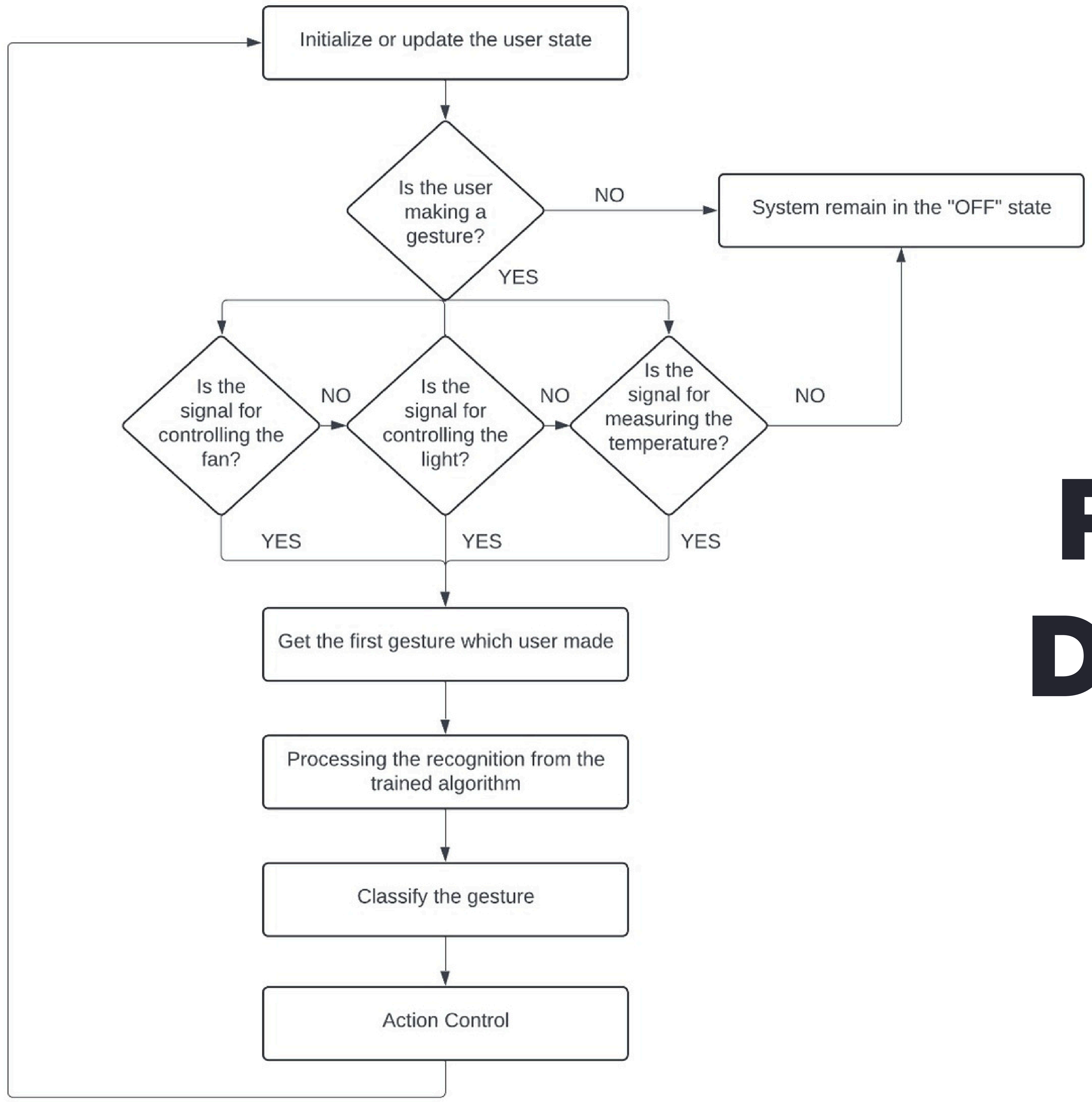
Device “OFF”

Level Up (FAN)

Level down (FAN)

Check Temperature

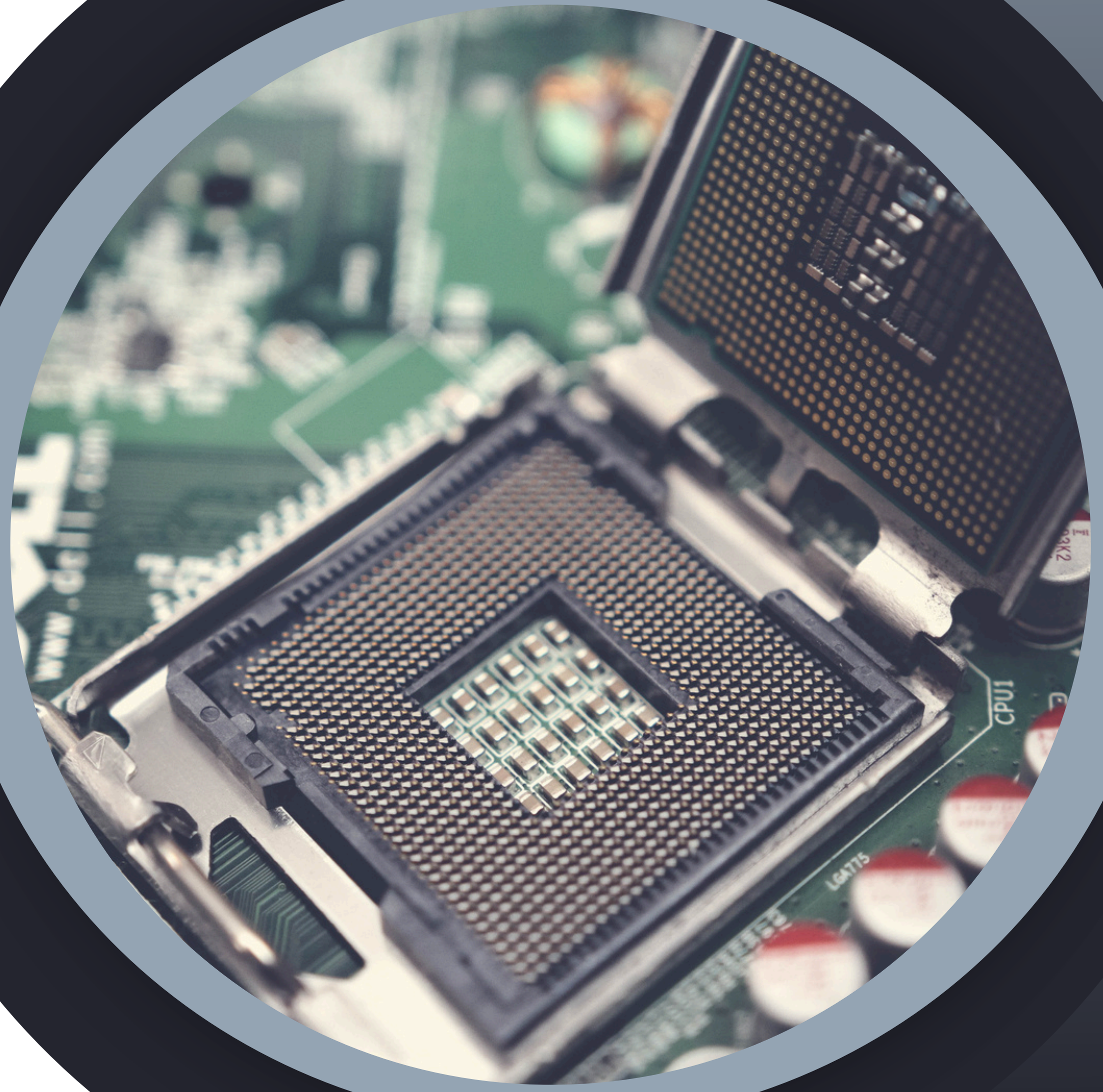
Design Diagram



Process Diagram

Future Development

1. **Smart Door Lock System** - The smart door locking system with hand gesture recognition allows users to lock and unlock doors using specific hand movements, eliminating the need for keys or keypads. It integrates a gesture recognition module, microcontroller, and smart lock mechanism for seamless operation. This setup enhances security and convenience, making it a modern and accessible solution for home security.
2. **Robotic Arm** - A robotic arm functions by using motors and sensors to move and manipulate objects in a controlled manner, mimicking human arm movements.
3. **Smart Virtual Temp Calculator & Display** - The Smart Virtual Temp Calculator & Display measures and calculates real-time temperature using sensors and a microcontroller, displaying the data on a screen for easy monitoring.



THANK YOU!

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